STAT 527 Advanced Regression Analysis
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R^2: Coefficient of Determination

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Goodness of Fit

It is often helpful in practice to provide a measure of how well our linear model explains the outcome variable (something in the range of [0,1] indicating worst to perfect performance).

It is of course too optimistic to hope that a single number can provide such a simple summary, but one commonly used measure that helps practitioners to get an idea about the model fit is R^2.

Coefficient of Determination R^2

Formally, the definition of R^2 is

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

where RSS is the sum of squared residuals, and TSS is the total sum of squares for Y: $TSS = \sum_{i=1}^{n}(y_i - \overline{y})^2$. Show that:

(When there is intercept)

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} = \frac{\sum_i (\hat{y}_i - \bar{y})^2}{\sum_i (y_i - \bar{y})^2}$$

R^2 is the fraction of TSS that is "explained" by the regression:

$$0 \leq R^2 \leq 1$$

where 0 represents no (linear) relationship, and 1 represents a perfect linear fit.

Coefficient of Determination R^2

An equivalent expression for R^2 is given by

$$R^2 = \hat{\text{cor}}(Y, \hat{Y})^2.$$

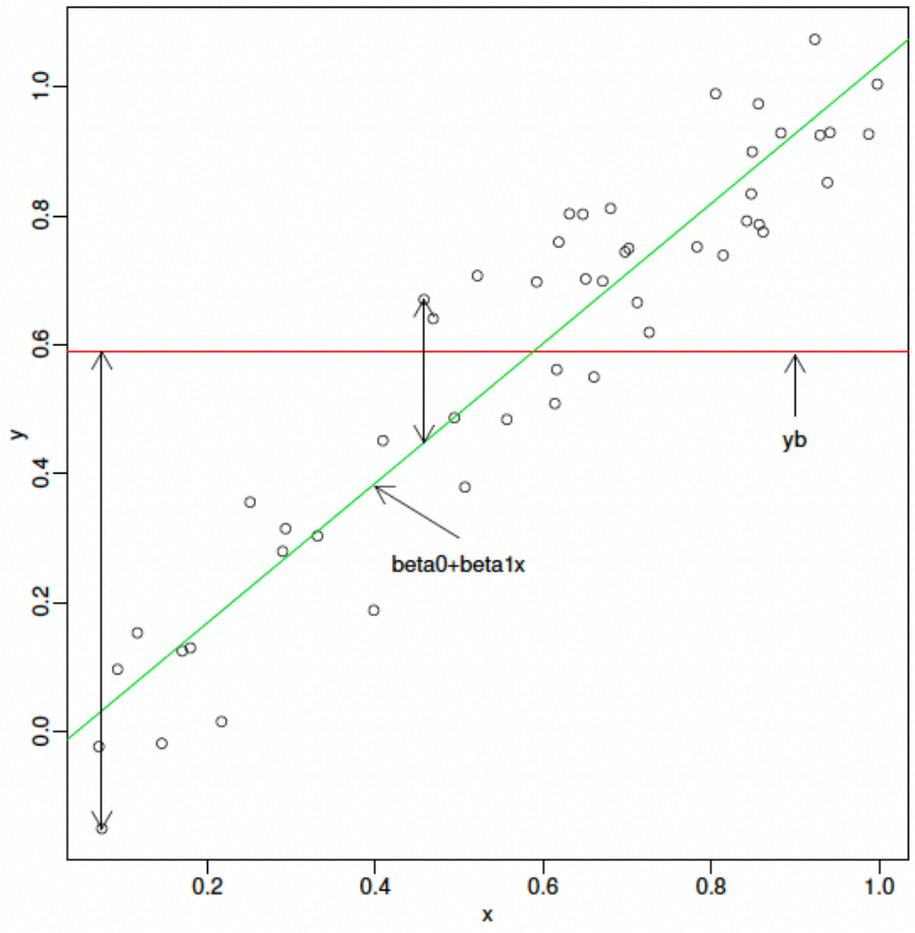
(When there is intercept)

Proof?

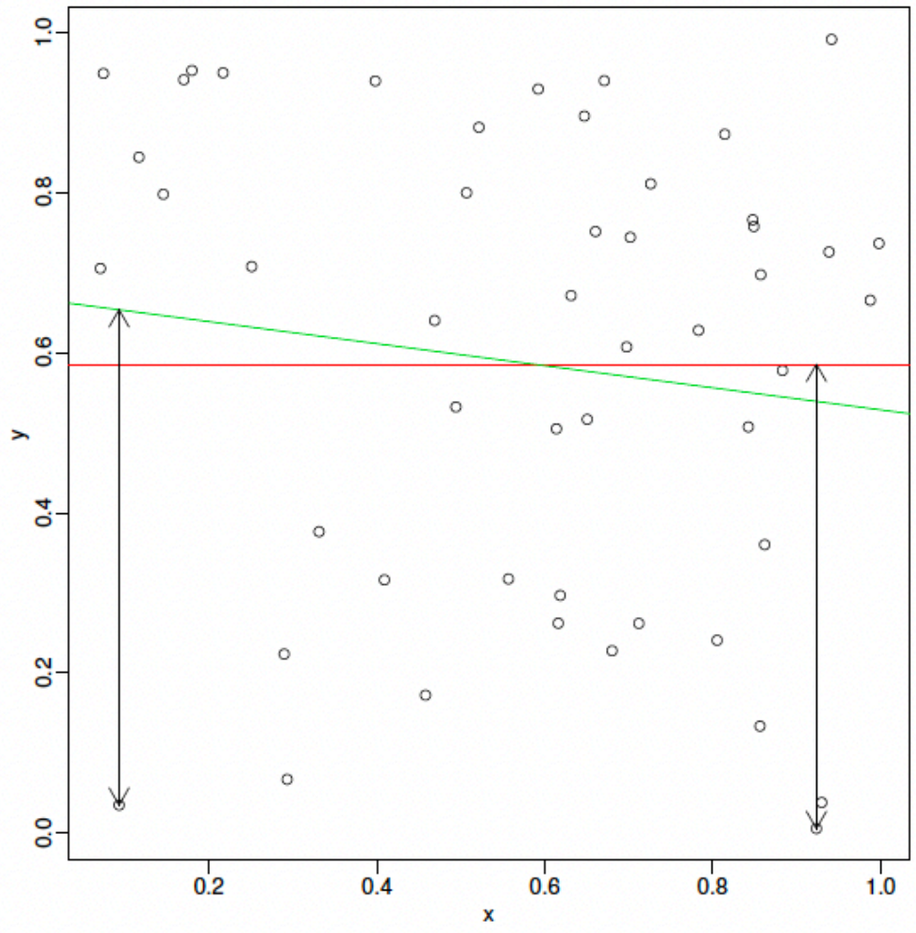
This explains that R^2 only measures the linear correlationship between Y and Y. It does NOT indicate whether:

- the regression model used is appropriate
- covariates have a substantial relationship with the outcome variable **(imagine a line passing 2 points)**
- the covariates are collinear
- there are enough observations to make a statistically valid conclusion

Intuition



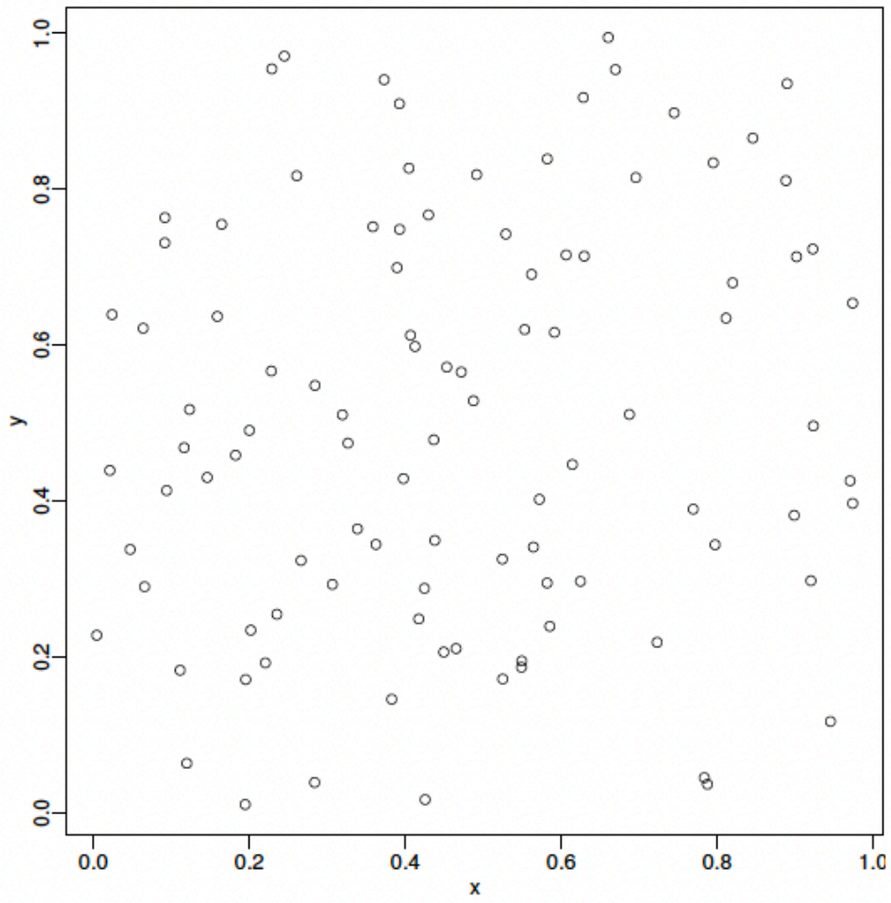
$R^2 = 0.89$

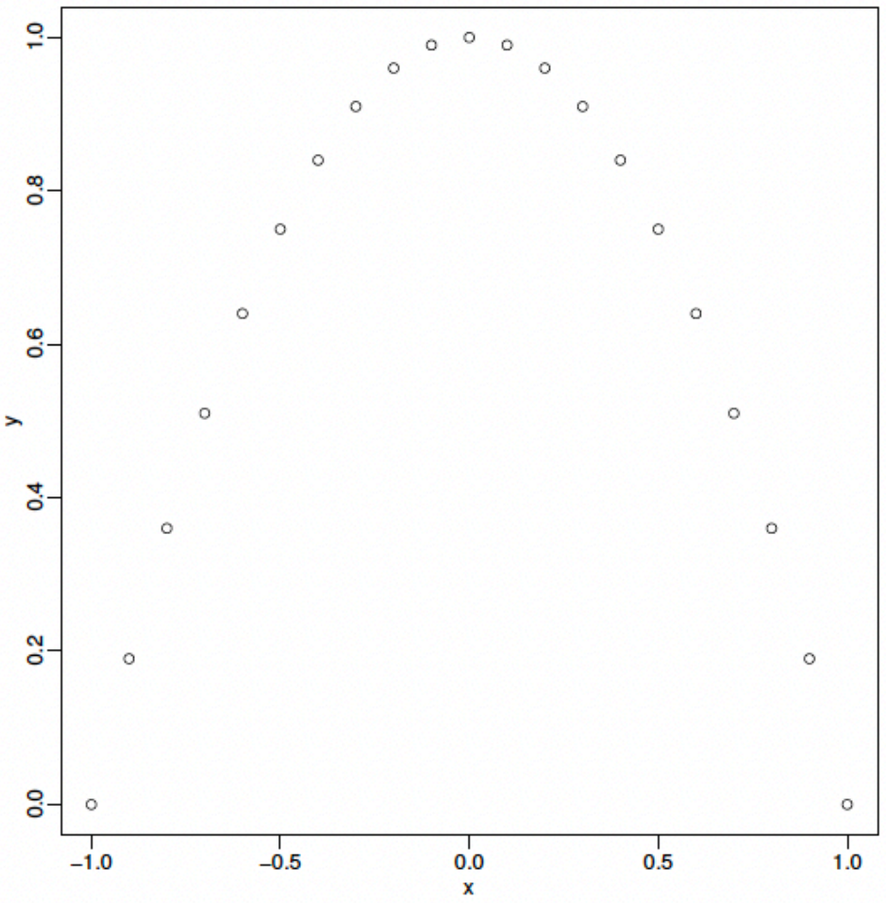


$R^2 = 0$

Remarks on R^2

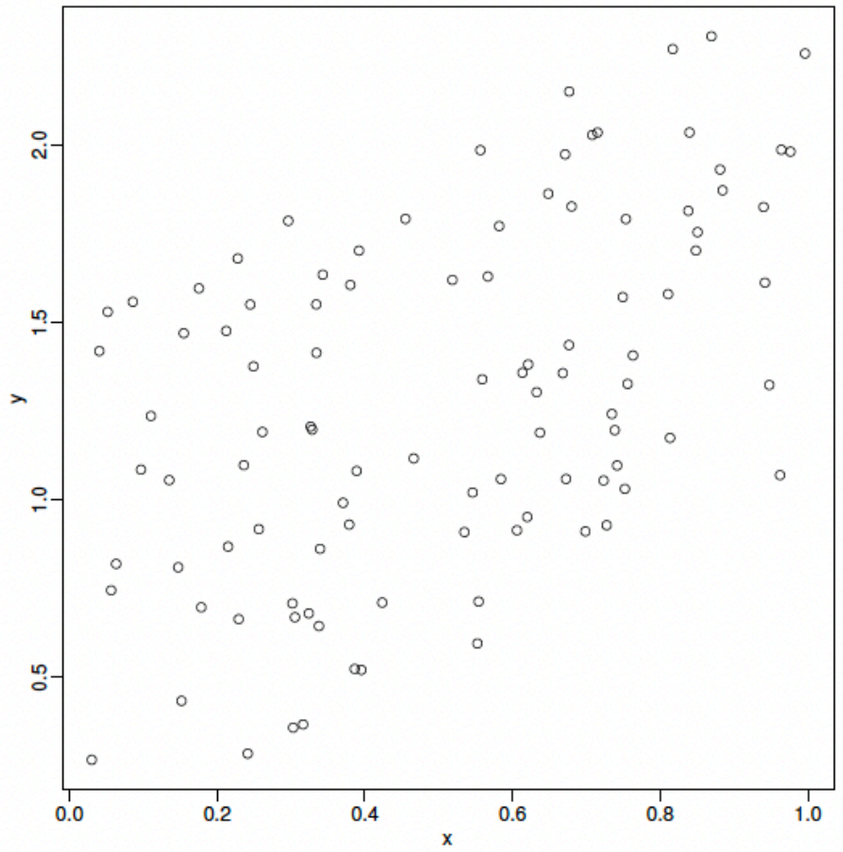
R^2 near 0 could be





Remarks on R^2 Continued

Small R^2 does not mean that y and X are not linearly related (can have slight trend with high variance).



$$R^2 = 0.2.$$

Caution: Small R^2 Does Imply Uncorrelatedness

$R^2 = 0$

$\Rightarrow y = \hat{y}$

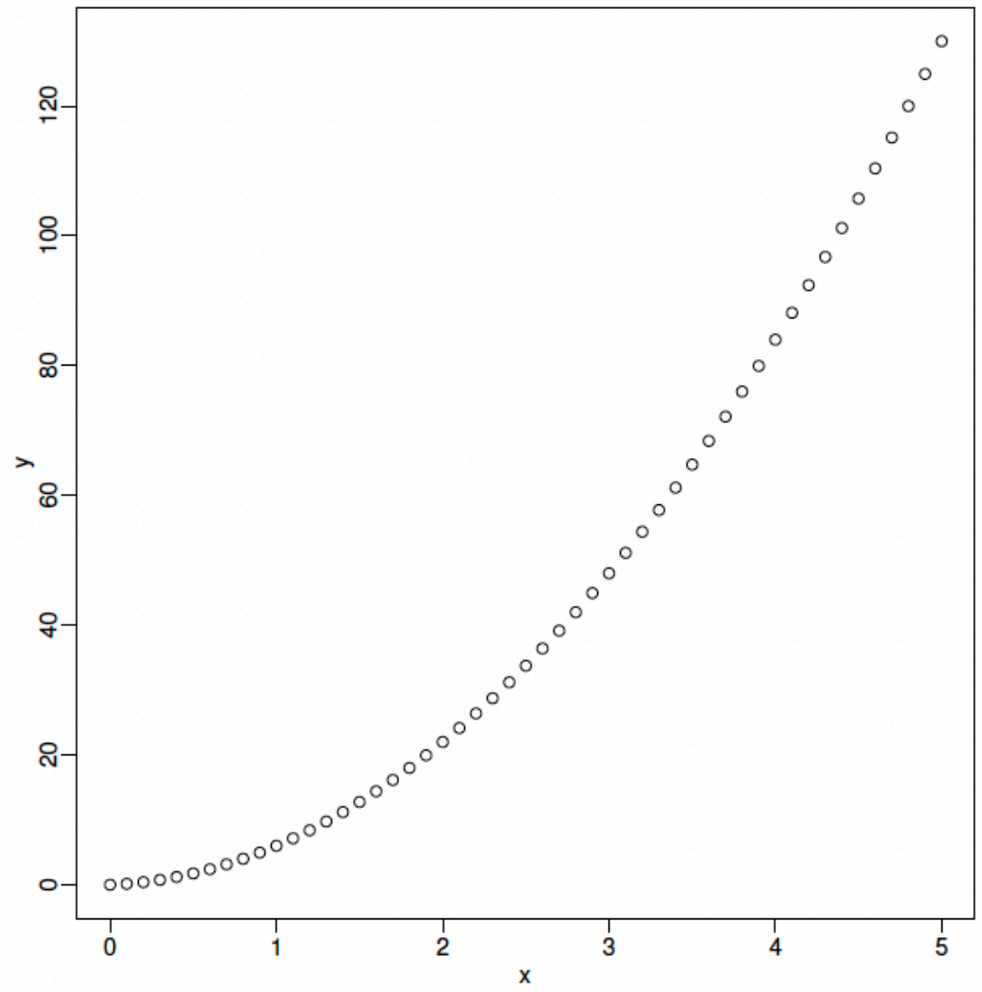
$\Rightarrow y - \bar{y} = y - \hat{y}$

they are perpendicular to $(1, \ldots, 1)^\top$ and $(x_1, \ldots, x_n)^\top$

$\Rightarrow y - \bar{y}$ and $x - \bar{x}$ are uncorrelated

Remarks on R^2 continued

- R^2 close to 1 does not mean the linear model is correct.



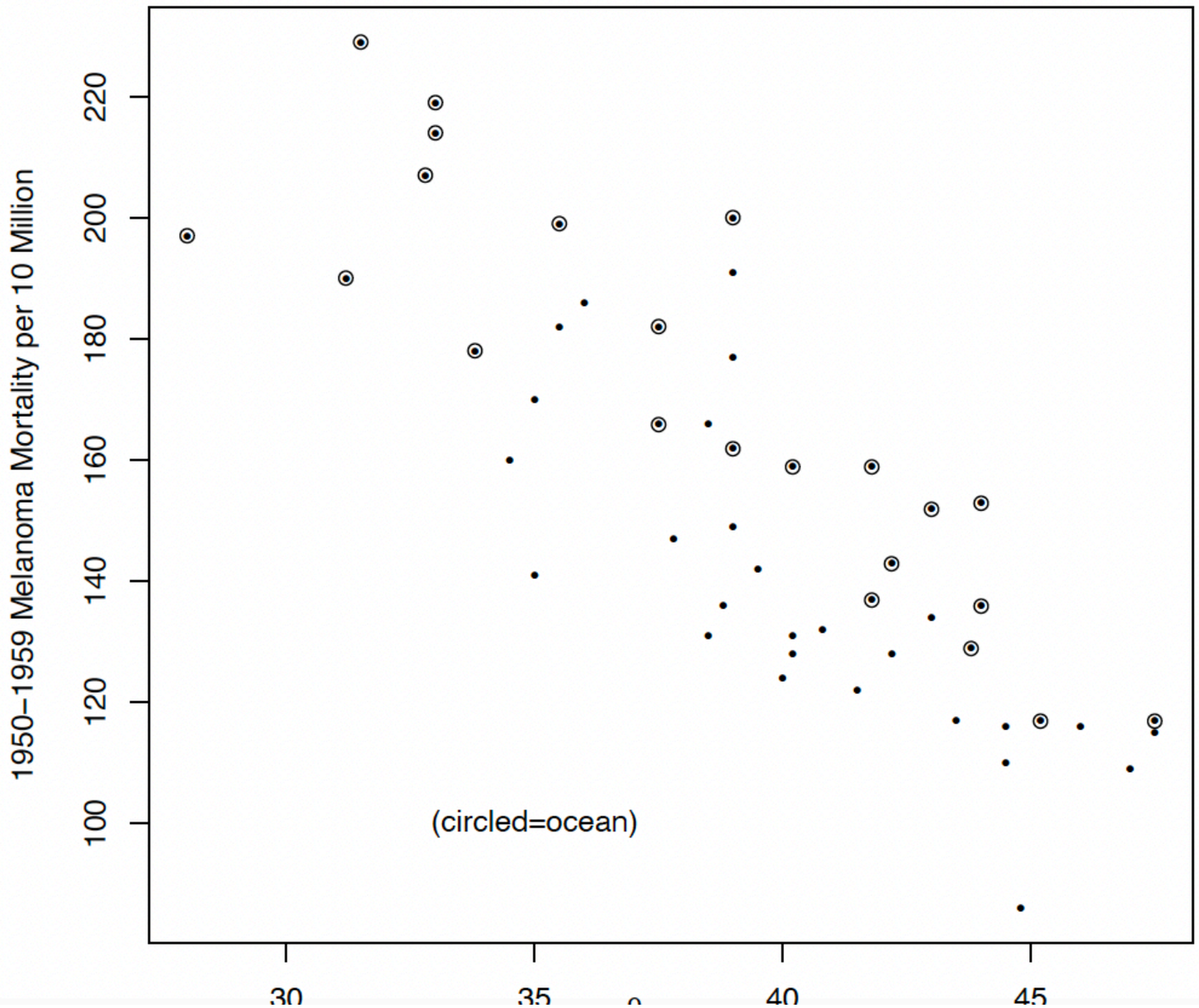
$$R^2 = 0.9.$$

Melanoma Data Revisited

Recall the melanoma mortality data ...

In addition to `latitude`, we will fit the predictor `ocean`, which is a **dummy variable** that takes the value 1 for states that border an ocean, and is 0 otherwise.

Melanoma Data Plot



Ocean as Predictor

Let's try ocean by itself first:

```
> summary(lm(mortality ~ ocean, data=Melanoma))

Call:
lm(formula = mortality ~ ocean, data = Melanoma)

Residuals:
    Min      1Q  Median      3Q     Max
-53.227 -21.741  -6.741  26.773  58.773

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  138.741      5.727  24.226  < 2e-16 ***
ocean         31.487      8.547   3.684 0.000592 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 29.76 on 47 degrees of freedom
Multiple R-squared:  0.2241,Adjusted R-squared:  0.2076
F-statistic: 13.57 on 1 and 47 DF,  p-value: 0.0005924
```

Latitude and Ocean as Predictors

Now with both `latitude` and `ocean`:

```
> summary(lm(mortality ~ latitude + ocean, data=Melanoma))

Call:
lm(formula = mortality ~ latitude + ocean, data = Melanoma)

Residuals:
    Min      1Q  Median      3Q     Max
-30.435 -11.328  -1.884   9.327  44.372

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 360.6905    21.4984  16.778  < 2e-16 ***
latitude     -5.4888     0.5259 -10.437 1.03e-13 ***
ocean        20.4304     4.8251   4.234 0.000109 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 16.39 on 46 degrees of freedom
Multiple R-squared:  0.7696,Adjusted R-squared:  0.7596
F-statistic: 76.83 on 2 and 46 DF,  p-value: 2.169e-15
```

Coefficient of Determination

Here we see that adding `ocean` increases R^2: It was 0.6798 for the model with only `latitude`, and it is 0.7696 for the model with both.

Because the correlation between `ocean` and `latitude` is not exactly 0, the R^2 for the multiple regression model is *not* just the sum of the R^2's from the two simple linear regression models.

Exercise: What if the Features are Uncorrelated?

If the two features x^o and x^l are uncorrelated (i.e. $x^o - \bar{x}^o \perp x^l - \bar{x}^l$) then $R^2 = R_o^2 + R_l^2$. To see this:

- Recall $R^2 := \frac{\|\hat{y} - \bar{y}\|^2}{\|y - \bar{y}\|^2}$;
- By Pythagorean's theorem, $\|\hat{y} - \bar{y}\|^2 = \|\hat{y}^o - \bar{y}\|^2 + \|\hat{y}^l - \bar{y}\|^2$;
- Dividing both sides by $\|y - \bar{y}\|^2$ yields the claim.

Coefficient of determination in simple linear regression

$$R^2 = \hat{\text{cor}}(Y, \hat{Y})^2.$$

In the case of simple linear regression,

$$\hat{\text{cov}}(Y; Y) = \hat{\text{cov}}(Y; \hat{\alpha} + \beta X) = \beta \hat{\text{cov}}(Y; X);$$

and

$$\hat{\text{var}}(\hat{Y}) = \hat{\text{var}}(\hat{\alpha} + \hat{\beta} X) = \hat{\beta}^2 \hat{\text{var}}(X);$$

Thus for simple linear regression,

$$R^2 = \hat{\text{cor}}(Y, \hat{Y})^2 = \hat{\text{cor}}(Y, X)^2.$$

Adjusted R^2

The sample R^2 is an estimate of the population version of R^2:

$$1 - \frac{\mathsf{var}(Y \mid X)}{\mathsf{var}(Y)}.$$

Since this involves a ratio, the plug-in estimate R^2 has a bias.

The adjusted R^2 is an approximately unbiased estimate of the population version of R^2:

$$R_a^2 := 1 - (1 - R^2)\frac{n-1}{n-p-1}$$

R_a^2 is always less than the unadjusted R^2. R_a^2 is always less than or equal to one, but can be negative.

Comments

- $R_a^2 = 1 - \frac{\frac{\| y - \hat{y} \|^2}{n - p'}}{\frac{\| y - \bar{y} \|^2}{n - 1}}$
- $R_a^2 \leq R^2 \leq 1$
- R^2 always improves by adding covariates, but R_a^2 does not and may be used for variable selection.